Multivariate Analysis of the Physicochemical Characteristics of Portuguese Vinho Verde Wines

1 Introduction

The chemical properties of wine affect its taste, quality, and overall characteristics. Properties such as acidity, sulfur dioxide, alcohol, and residual sugar can help explain the differences between wines. Studying these variables together can provide a better understanding of how they are related to one another. This study uses multivariate statistical methods to examine the chemical properties of Portuguese Vinho Verde wines and determine how red and white wines differ in terms of their chemical composition.

2 Dataset

This study uses the Wine Quality Dataset from the UCI Machine Learning Repository. The combined dataset contains 6,497 observations, consisting of 1,599 red wine samples and 4,898 white wine samples. It includes 11 continuous physicochemical variables: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. It also includes wine color as a categorical variable and a numerical quality score based on sensory evaluation.

3 Research Questions and Hypotheses

Research Question 1 (MANOVA): Is there a significant difference between red and white wines when their physicochemical characteristics are considered together?

- H_0: Red and white wines have the same mean values for fixed acidity, volatile acidity, residual sugar, and chlorides when considered as a group.
- H_1: Red and white wines have different mean values for at least one of the selected physicochemical characteristics.

Research Question 2 (PCA): Can the 11 physicochemical variables be summarized into a smaller number of principal components while retaining a large portion of the total variation in the data?

Research Question 3 (EFA): What underlying factors can explain the relationships among the physicochemical characteristics of wine?

4 Methodology

Before conducting the analysis, the red and white wine datasets were combined, the column names were standardized, and exact duplicate observations were removed. Exploratory Data Analysis (EDA) was then performed using boxplots, density plots, and correlation matrices to examine the distributions of the variables and identify relationships between them.

For the first research question, a one-way MANOVA was conducted using wine color as the independent variable and the four selected chemical properties as dependent variables. Since Box's M test indicated that the covariance matrices were unequal, Pillai's Trace was used because it is generally more reliable when this assumption is violated.

Principal Component Analysis (PCA) was performed on the standardized values of all 11 physicochemical variables to identify a smaller number of components that could explain much of the variation in the data. Exploratory Factor Analysis (EFA) was also conducted using Principal Axis Factoring and Oblimin rotation. This rotation was used because the underlying factors may be related to one another. All analyses were performed in R using packages such as `tidyverse`, `psych`, and `factoextra`.

5 Results

5.1 Exploratory Data Analysis

The initial analysis showed differences between red and white wines. Red wines generally had higher volatile acidity and chloride levels, while white wines had higher sulfur dioxide levels and residual sugar. The sulfur dioxide variables also showed relatively high variation across the samples.

The correlation matrix showed some strong relationships between variables. Density and alcohol had a negative correlation ($r = -0.67$), meaning that wines with higher alcohol levels tended to have lower density. Meanwhile, free and total sulfur dioxide had a positive correlation ($r = 0.72$). These relationships suggest that some variables are closely related, so standardization was applied before conducting PCA and EFA.

5.2 MANOVA

The MANOVA results showed a significant difference between red and white wines based on the four selected physicochemical variables. Pillai's Trace was 0.631, with $F(4, 6492) = 2775.6$ and $p < 0.001$. The partial eta-squared was also large ($\eta_p^2 = 0.631$), indicating a strong difference between the two wine groups in this analysis.

The follow-up univariate ANOVAs showed that all four variables differed significantly between red and white wines ($p < 0.001$ for all variables). This suggests that wine color is associated with differences in acidity, residual sugar, and chloride levels. However, these results show an association between wine color and chemical composition rather than proving that wine color directly causes these differences.

5.3 PCA

Based on the eigenvalue-greater-than-one rule and the scree plot, three principal components were retained. Together, they explained 64.36% of the total variation in the 11 physicochemical variables.

- **PC1 (27.5% of variance):** This component was mainly associated with sulfur dioxide and volatile acidity. It can be interpreted as a general sulfur dioxide and acidity profile.
- **PC2 (22.7% of variance):** This component was mainly associated with density and alcohol. Since these variables had opposite loadings, it can be interpreted as a density and alcohol profile.
- **PC3 (14.1% of variance):** This component was associated with citric acid, fixed acidity, and pH. It may represent differences in the acidity characteristics of the wines.

Overall, PCA reduced the 11 original variables into three components while retaining a considerable amount of information. However, 35.64% of the total variation was still not explained by these components.

5.4 EFA

Parallel analysis suggested retaining three factors. These factors represented groups of variables that were related to one another.

- **Factor 1 (Sulfite and Sweetness):** This factor was mainly associated with sulfur dioxide and residual sugar. It may represent the relationship between sugar content and sulfur dioxide levels in the wines.
- **Factor 2 (Density and Alcohol):** This factor represented the relationship between density and alcohol content. These variables were strongly related but had opposite directions in their loadings. The analysis also produced an ultra-Heywood case, which suggests that this factor solution may have estimation problems and should be interpreted with caution.
- **Factor 3 (Acidity and pH):** This factor was mainly associated with citric acid and fixed acidity, along with pH. It may represent differences in the acidity of the wines.

Although the three-factor solution provided a possible way to summarize the relationships among the variables, the results should be interpreted carefully. The low overall Kaiser-Meyer-Olkin (KMO) value of 0.4 suggests that the data may not be well suited for factor analysis.

6 Integrated Discussion

The results of MANOVA, PCA, and EFA provide different but related insights into the chemical characteristics of Vinho Verde wines. MANOVA showed that red and white wines differ significantly based on the four selected chemical variables. PCA summarized the 11 physicochemical variables into three main components related to sulfur dioxide and acidity, density and alcohol, and other acidity characteristics.

EFA also suggested three possible underlying factors, although its results were less reliable because of the low KMO value and the ultra-Heywood case. The findings from PCA and EFA show some similarities, particularly in the relationships among sulfur dioxide, residual sugar, density, alcohol, and acidity. However, the components identified by PCA and the factors identified by EFA are not exactly the same because the two methods have different purposes.

One limitation of the study is that the dataset contains more white wine samples than red wine samples. Another limitation is the low KMO value, which suggests that the relationships among the variables may not be strong enough to support a reliable factor analysis. These limitations should be considered when interpreting the results, especially the factors identified through EFA.

7 Conclusion

This study used MANOVA, PCA, and EFA to examine the physicochemical characteristics of Portuguese Vinho Verde wines. The MANOVA results showed significant differences between red and white wines based on fixed acidity, volatile acidity, residual sugar, and chlorides. PCA reduced the 11 physicochemical variables into three principal components that explained 64.36% of the total variation. EFA also suggested three factors related to sulfur dioxide and sweetness, density and alcohol, and acidity.

Overall, the results show that multivariate statistical methods can help explain the relationships among the chemical properties of wine and identify differences between red and white wines. However, the EFA results should be interpreted with caution because of the low KMO value and the estimation issue encountered. Further analysis may help determine whether a more reliable factor structure can be obtained.

8 References

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